

A Sufficient-Condition Analysis of SENTINEL-SplitFed Staleness-Normalized Encoder Anomaly Detection for Intrusion-Resilient Asynchronous Split Federated Learning

Proof note for manuscript integration

Abstract

This note gives a formal sufficient-condition argument for the SENTINEL-SplitFed mechanism. The analysis does not claim an unconditional Byzantine guarantee: no memoryless anomaly detector can guarantee detection against all adaptive adversaries without assumptions on the separation between honest drift and malicious perturbation. Instead, we prove that when honest staleness-induced drift is bounded and malicious encoder submissions are sufficiently separated after staleness normalization, the Staleness-Normalized Anomaly Score (SNAS) accepts honest stragglers, quarantines malicious submissions, and makes the asynchronous aggregate a convex combination of bounded honest updates. We also prove a simple structural limitation: an attack-then-abstain adversary can evade any per-round staleness-normalized gate when the staleness denominator suppresses the anomaly below the decision threshold. This matches the empirical robustness boundary observed in SENTINEL-SplitFed.

1 Setting

Consider an asynchronous split federated learning system with clients $i \in \{1, \dots, N\}$. Client i holds a local encoder f_{θ_i} with parameter vector $\theta_i \in \mathbb{R}^d$, while the server holds the top model. At aggregation round r , the previous global encoder is θ_r^g . A client may submit an encoder $\theta_i(r)$ within the timed window, or miss the window. The staleness counter is

$$\tau_i(r) = \text{number of consecutive missed aggregation rounds before round } r. \quad (1)$$

If client i submits in round r , the server computes an anomaly score relative to θ_r^g .

For notational simplicity, write the encoder as K tensors $\theta_{i,1}, \dots, \theta_{i,K}$. Define

$$A_i^{\text{cos}}(r) = 1 - \frac{1}{K} \sum_{k=1}^K \cos\left(\text{vec}(\theta_{i,k}(r)), \text{vec}(\theta_{r,k}^g)\right), \quad (2)$$

$$A_i^{\text{norm}}(r) = \frac{1}{K} \sum_{k=1}^K \min \left\{ \frac{\left| \log \left(\frac{\|\theta_{i,k}(r)\|_2}{\|\theta_{r,k}^g\|_2 + \epsilon} \right) \right|}{\log 100}, 1 \right\}. \quad (3)$$

The unnormalized evidence is

$$S_i(r) = \beta A_i^{\text{cos}}(r) + \gamma A_i^{\text{norm}}(r), \quad \beta, \gamma \geq 0. \quad (4)$$

The Staleness-Normalized Anomaly Score is

$$\text{SNAS}_i(r) = \frac{S_i(r)}{1 + \tau_i(r)}. \quad (5)$$

The decision rule uses thresholds $0 < \theta_f < \theta_q$:

$$\text{gate}_i(r) = \begin{cases} \text{accept}, & \text{SNAS}_i(r) \leq \theta_f, \\ \text{flag}, & \theta_f < \text{SNAS}_i(r) \leq \theta_q, \\ \text{quarantine}, & \text{SNAS}_i(r) > \theta_q. \end{cases} \quad (6)$$

Flagged clients are downweighted; quarantined clients are excluded.

2 Assumptions for a Positive Guarantee

The following assumptions are intentionally explicit. They state the regime in which a staleness-normalized detector should work. They are not automatic consequences of the algorithm.

Assumption 1 (Honest staleness-adjusted bounded drift). *For every honest client i that submits in round r ,*

$$S_i(r) \leq (1 + \tau_i(r))\theta_f. \quad (7)$$

Equivalently, honest update drift, including delay-induced drift, remains below the acceptance threshold after staleness normalization.

Assumption 2 (Malicious separation). *For every malicious client m that submits in round r ,*

$$S_m(r) > (1 + \tau_m(r))\theta_q. \quad (8)$$

Equivalently, malicious encoder evidence remains above the quarantine threshold even after staleness normalization.

Assumption 3 (At least one usable honest submission). *In every aggregation round considered, at least one honest client submits within the window.*

Assumption 4 (Bounded honest update radius). *There exists $\delta_r \geq 0$ such that every honest submitted encoder in round r satisfies*

$$\|\theta_i(r) - \theta_r^g\|_2 \leq \delta_r. \quad (9)$$

3 Detection Correctness

Lemma 1 (No false quarantine for honest submissions). *Under Assumption 1, every honest submitted client is accepted by the SNAS gate.*

Proof. For an honest submitted client i , Assumption 1 gives

$$S_i(r) \leq (1 + \tau_i(r))\theta_f. \quad (10)$$

Dividing both sides by $1 + \tau_i(r) > 0$ yields

$$\text{SNAS}_i(r) = \frac{S_i(r)}{1 + \tau_i(r)} \leq \theta_f. \quad (11)$$

By the gate definition, client i is accepted. Hence no honest submitted client is falsely flagged or quarantined in this sufficient-condition regime. $\square$

Lemma 2 (Quarantine of separated malicious submissions). *Under Assumption 2, every malicious submitted client is quarantined by the SNAS gate.*

Proof. For a malicious submitted client m , Assumption 2 gives

$$S_m(r) > (1 + \tau_m(r))\theta_q. \quad (12)$$

Dividing by $1 + \tau_m(r) > 0$ yields

$$\text{SNAS}_m(r) = \frac{S_m(r)}{1 + \tau_m(r)} > \theta_q. \quad (13)$$

By the gate definition, client m is quarantined. $\square$

Corollary 1 (Perfect gate separation under the assumptions). *If Assumptions 1 and 2 hold in round r , then the SNAS gate accepts all honest submitted clients and quarantines all malicious submitted clients in that round.*

4 Aggregation Stability

For a non-quarantined client i , SENTINEL-SplitFed assigns raw aggregation weight

$$\omega_i(r) = n_i d(\tau_i(r)) p_i(r), \quad (14)$$

where $n_i > 0$ is the local sample count, $d(\tau_i) > 0$ is a staleness decay, and $p_i \in (0, 1]$ is a gate penalty. The normalized weight is

$$\bar{\omega}_i(r) = \frac{\omega_i(r)}{\sum_{j \in \mathcal{S}_r} \omega_j(r)}, \quad (15)$$

where $\mathcal{S}_r$ is the set of accepted or flagged clients. The aggregate is

$$\theta_{r+1}^g = \sum_{i \in \mathcal{S}_r} \bar{\omega}_i(r) \theta_i(r). \quad (16)$$

Lemma 3 (Convexity of the accepted aggregate). *If $\mathcal{S}_r$ is nonempty and $\omega_i(r) > 0$ for all $i \in \mathcal{S}_r$, then θ_{r+1}^g is a convex combination of the non-quarantined submitted encoders.*

Proof. By construction, $\bar{\omega}_i(r) > 0$ and

$$\sum_{i \in \mathcal{S}_r} \bar{\omega}_i(r) = \sum_{i \in \mathcal{S}_r} \frac{\omega_i(r)}{\sum_{j \in \mathcal{S}_r} \omega_j(r)} = 1. \quad (17)$$

Therefore the aggregate is a convex combination of the submitted encoders indexed by $\mathcal{S}_r$. $\square$

Lemma 4 (Bounded honest aggregate drift). *Assume Corollary 1, Assumption 3, and Assumption 4. Then*

$$\|\theta_{r+1}^g - \theta_r^g\|_2 \leq \delta_r. \quad (18)$$

Proof. By Corollary 1, all malicious submissions are quarantined and all honest submitted clients are accepted. By Assumption 3, the aggregation set is nonempty and contains only honest clients. Therefore,

$$\|\theta_{r+1}^g - \theta_r^g\|_2 = \left\| \sum_{i \in \mathcal{S}_r} \bar{\omega}_i(r) \theta_i(r) - \theta_r^g \right\|_2 \quad (19)$$

$$= \left\| \sum_{i \in \mathcal{S}_r} \bar{\omega}_i(r) (\theta_i(r) - \theta_r^g) \right\|_2 \quad (20)$$

$$\leq \sum_{i \in \mathcal{S}_r} \bar{\omega}_i(r) \|\theta_i(r) - \theta_r^g\|_2 \quad (21)$$

$$\leq \sum_{i \in \mathcal{S}_r} \bar{\omega}_i(r) \delta_r = \delta_r. \quad (22)$$

The first inequality is the triangle inequality, and the final equality uses the fact that the normalized weights sum to one. $\square$

Lemma 5 (Norm clipping bound). *For each tensor k , the clipping rule*

$$\tilde{\theta}_{i,k} = \theta_{i,k} \cdot \min \left\{ 1, \frac{c_{\text{clip}} \|\theta_{r,k}^g\|_2}{\|\theta_{i,k}\|_2 + \epsilon} \right\} \quad (23)$$

implies

$$\|\tilde{\theta}_{i,k}\|_2 \leq c_{\text{clip}} \|\theta_{r,k}^g\|_2. \quad (24)$$

Consequently, any convex aggregate of clipped submitted tensors also satisfies

$$\|\tilde{\theta}_{r+1,k}^g\|_2 \leq c_{\text{clip}} \|\theta_{r,k}^g\|_2. \quad (25)$$

Proof. If the minimum equals 1, then the condition for this case implies

$$\|\theta_{i,k}\|_2 \leq c_{\text{clip}} \|\theta_{r,k}^g\|_2 + \epsilon. \quad (26)$$

Ignoring the arbitrarily small numerical stabilizer ϵ gives the stated bound; with ϵ retained, the bound holds up to an ϵ tolerance. If the minimum equals the second term, then

$$\|\tilde{\theta}_{i,k}\|_2 = \|\theta_{i,k}\|_2 \frac{c_{\text{clip}} \|\theta_{r,k}^g\|_2}{\|\theta_{i,k}\|_2 + \epsilon} \leq c_{\text{clip}} \|\theta_{r,k}^g\|_2. \quad (27)$$

For the aggregate, convexity and the triangle inequality give

$$\left\| \sum_i \bar{\omega}_i \tilde{\theta}_{i,k} \right\|_2 \leq \sum_i \bar{\omega}_i \|\tilde{\theta}_{i,k}\|_2 \leq c_{\text{clip}} \|\theta_{r,k}^g\|_2 \sum_i \bar{\omega}_i = c_{\text{clip}} \|\theta_{r,k}^g\|_2. \quad (28)$$

$\square$

5 Main Sufficient-Condition Theorem

Theorem 1 (SENTINEL-SplitFed works under staleness-adjusted separation). *Suppose that, in every considered round r , Assumptions 1–4 hold. Then SENTINEL-SplitFed satisfies the following properties.*

- (i) *Every honest submitted client is accepted by the SNAS gate.*
- (ii) *Every malicious submitted client satisfying the malicious separation condition is quarantined.*
- (iii) *The aggregation set contains only honest submitted encoders.*
- (iv) *The new global encoder is a convex combination of honest submitted encoders.*
- (v) *The global encoder drift is bounded as $\|\theta_{r+1}^g - \theta_r^g\|_2 \leq \delta_r$.*
- (vi) *With norm clipping, each aggregated tensor norm is bounded by $c_{\text{clip}} \|\theta_{r,k}^g\|_2$ up to numerical tolerance.*

Proof. Parts (i) and (ii) follow from Lemma 1 and Lemma 2. Part (iii) follows because all malicious submitted clients are quarantined and therefore excluded. Part (iv) follows from Lemma 3. Part (v) follows from Lemma 4. Part (vi) follows from Lemma 5. $\square$

Corollary 2 (Bounded loss degradation under a Lipschitz condition). *Let $\mathcal{L}(\theta)$ denote the validation loss of the SplitFed model as a function of the client encoder parameters with the server top model fixed in the aggregation step. If $\mathcal{L}$ is L -Lipschitz in the encoder parameters around θ_r^g , then under Theorem 1,*

$$|\mathcal{L}(\theta_{r+1}^g) - \mathcal{L}(\theta_r^g)| \leq L\delta_r. \quad (29)$$

Proof. By L -Lipschitz continuity,

$$|\mathcal{L}(\theta_{r+1}^g) - \mathcal{L}(\theta_r^g)| \leq L\|\theta_{r+1}^g - \theta_r^g\|_2. \quad (30)$$

Applying Lemma 4 gives the result. $\square$

6 Why the Staleness Term Helps Honest Stragglers

The core design idea is that honest delay-induced drift should be discounted. Let $S_i(r)$ be the raw anomaly evidence for an honest stale client. Without staleness normalization, the client would be accepted only if $S_i(r) \leq \theta_f$. With SNAS, it is accepted if

$$\frac{S_i(r)}{1 + \tau_i(r)} \leq \theta_f, \quad (31)$$

which is equivalent to

$$S_i(r) \leq (1 + \tau_i(r))\theta_f. \quad (32)$$

Thus the tolerated honest raw drift grows linearly with staleness. This is useful when delay causes benign drift away from the current global reference.

For an on-time malicious client, $\tau_m = 0$, so the detector applies no staleness discount:

$$\text{SNAS}_m(r) = S_m(r). \quad (33)$$

Therefore an on-time attack with $S_m(r) > \theta_q$ is quarantined immediately after warm-up.

7 A Formal Vulnerability of Memoryless Staleness Normalization

The positive theorem requires malicious separation after staleness normalization. The following proposition shows why an adaptive attack-then-abstain adversary can violate this condition even when its raw anomaly would be detected at zero staleness.

Proposition 1 (Attack-then-abstain evasion condition). *Consider a malicious client with raw anomaly evidence S_m . Suppose that the client deliberately misses one round before submitting, so that $\tau_m = 1$. Then:*

(a) *The attack is accepted if $S_m \leq 2\theta_f$.*

(b) *The attack avoids quarantine if $S_m \leq 2\theta_q$.*

(c) *There exists a nonempty interval of raw anomalies that are detectable at $\tau = 0$ but accepted at $\tau = 1$, namely*

$$\theta_f < S_m \leq 2\theta_f. \quad (34)$$

(d) *There exists a nonempty interval of raw anomalies that are quarantined at $\tau = 0$ but not quarantined at $\tau = 1$, namely*

$$\theta_q < S_m \leq 2\theta_q. \quad (35)$$

Proof. If $\tau_m = 1$, then

$$\text{SNAS}_m = \frac{S_m}{2}. \quad (36)$$

The attack is accepted exactly when $S_m/2 \leq \theta_f$, which is equivalent to $S_m \leq 2\theta_f$. It avoids quarantine exactly when $S_m/2 \leq \theta_q$, which is equivalent to $S_m \leq 2\theta_q$. At $\tau = 0$, the same raw anomaly has $\text{SNAS}_m = S_m$. Thus any $S_m > \theta_f$ would not be accepted at zero staleness, but if $S_m \leq 2\theta_f$ it is accepted after one deliberate missed round. Similarly, any $S_m > \theta_q$ would be quarantined at zero staleness, but if $S_m \leq 2\theta_q$ it is not quarantined after one deliberate missed round. Since $\theta_f > 0$ and $\theta_q > 0$, both intervals are nonempty. $\square$

Remark 1 (Implication). *Proposition 1 proves that memoryless staleness normalization creates an evasion window. This does not contradict Theorem 1; instead, it shows exactly when Assumption 2 fails. A mitigation is to accumulate anomaly evidence over time, for example*

$$H_i(r) = \lambda H_i(r-1) + (1-\lambda) \text{SNAS}_i(r), \quad 0 < \lambda < 1, \quad (37)$$

or to cap the staleness denominator as $1 + \min\{\tau_i, \tau_{\max}\}$, or to dampen only the components of the score that are plausibly caused by benign delay.

8 Manuscript-Ready Statement

The method can be described with the following theorem-level claim:

SENTINEL-SplitFed provides a conditional robustness guarantee: if honest staleness-induced encoder drift remains below the staleness-adjusted flag threshold and malicious encoder perturbations remain above the staleness-adjusted quarantine threshold, then the SNAS gate accepts honest submissions, quarantines malicious submissions, and reduces asynchronous aggregation to a bounded convex combination of honest encoders. Norm clipping further bounds the magnitude of any submitted tensor during warm-up. The guarantee is not unconditional; adaptive participation manipulation can violate the malicious-separation condition, motivating temporal anomaly accumulation.

9 Conclusion

This proof clarifies what can and cannot be claimed mathematically. SENTINEL-SplitFed is provably correct as a detection-and-aggregation mechanism under explicit staleness-adjusted separation conditions. The proof supports the empirical findings that honest stragglers should not be falsely quarantined and that sufficiently strong gradient-scaling or free-rider attacks should be removed from aggregation. It also formally explains the observed attack-then-abstain weakness: the same denominator that protects honest stale clients can be exploited by adaptive malicious clients unless anomaly evidence is accumulated across rounds or the staleness discount is capped.